

Date: 85/12/2023

Science

Physics

1. (3) Concave Mirror

2.

$$3 = \frac{h_2}{h_1}$$

$$\Rightarrow 3h_1 = h_2$$

$$+v = \frac{h_2}{h_1}$$

$$\Rightarrow \frac{v}{-10} = \frac{3h_1}{h_1}$$

$$\Rightarrow v = 30 \text{ cm}$$

$$\frac{1}{u} = \frac{1}{v} - \frac{1}{f}$$

$$\frac{1}{6} = \frac{1}{30} - \frac{1}{f}$$

$$= \frac{1}{30} - \frac{3}{30}$$

$$\frac{1}{f} = \frac{4}{30}$$

$$\Rightarrow f = \frac{30}{4} = 7.5$$

Ans-Option (3) 30 cm, 7.5 cm

3. (4) Both 2 and 3

4. (i) Focus is the point on the principal axis in which all the ~~parallel rays~~ converge parallel to principal axis converge into a point.

(ii) ~~Centre of Curvature of spherical mirror is the point on the principal axis which the spherical mirror is cut into.~~

(ii) Centre of Curvature is the point on which centre of a ~~spherical~~ spherical mirror on which the spherical mirror is cut.

5 Given: $v = 18 \text{ cm}$

$$h_1 = 5h_2$$

$$h_2 = \frac{1}{5}h_1$$

$$(a) h_1 = 5h_2$$

$$\frac{v}{u} = \frac{h_2}{h_1}$$

$$\Rightarrow \frac{18}{u} = \frac{5h_2}{5h_1}$$

$$\Rightarrow -1 = 90 \text{ cm}$$

$$(a) u = -90 \text{ cm}$$

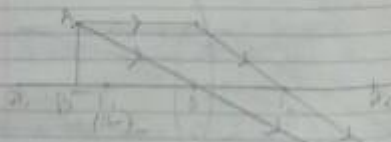
$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{18} - \frac{1}{-90}$$

$$= \frac{5+1}{90}$$

$$\frac{1}{f} = \frac{6}{90} \Rightarrow \frac{1}{f} = \frac{1}{15}$$

$$(a) f = 15 \text{ cm}$$



Q. (i) Given: $h_1 = 7 \text{ cm}$
 $u = -27 \text{ cm}$
 $f = -18 \text{ cm}$

To find: (a) v
 (b) h_2 & Nature of Image.

(a) $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$
 $\frac{1}{v} = \frac{1}{u} - \frac{1}{f}$
 $= \frac{1}{-27} - \frac{1}{-18}$
 $= \frac{-2 + 3}{54}$
 $= \frac{1}{54}$

$\Rightarrow \boxed{v = -54 \text{ cm}} \quad \dots (1)$

$\frac{-v}{u} = \frac{h_2}{h_1}$
 $-\left(\frac{-54}{-27}\right) = \frac{h_2}{7}$
 $h_2 = -14 \text{ cm}$

Size of image is 18cm and the nature of image is real and inverted.

(i) Given: $R = 5\text{cm} \Rightarrow$ (Mirror is Concave)
 $f = 2.5\text{cm}$
 $u = -20\text{cm}$ (or) -200cm

To find: v , Nature & Size of Image

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{2.5} + \frac{1}{200}$$

$$= \frac{80 + 1}{200}$$

$$v = \frac{200 \times 200}{81}\text{cm}$$

$$v = 2.46\text{cm}$$

$$\begin{array}{r} 200 \\ \times 200 \\ \hline 40000 \\ \times 81 \\ \hline 32400 \\ \hline 81000 \end{array}$$

Nature of the image is Virtual and Erect

(ii) The mirror is Concave Mirror

It is used by dentists so that can get ~~an image~~ a magnified image of the teeth when the teeth's between focus and pole.